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Pause Giant AI Experiments: An Open Letter

We call on all AI labs to immediately pause for at least 6 months the training of AI systems more powerful than GPT-4.

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33709

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AI systems with human-competitive intelligence can pose profound risks to society and humanity, as shown by extensive research^[1] and acknowledged by top AI labs.^[2] As stated in the widely-endorsed [Asilomar AI Principles](#), *Advanced AI could represent a profound change in the history of life on Earth, and should be planned for and managed with commensurate care and resources*. Unfortunately, this level of planning and management is not happening, even though recent months have seen AI labs locked in an out-of-control race to develop and deploy ever more powerful digital minds that no one – not even their creators – can understand, predict, or reliably control.

Contemporary AI systems are now becoming human-competitive at general tasks,^[3] and we must ask ourselves: *Should* we let machines flood our information channels with propaganda and untruth? *Should* we automate away all the jobs, including the fulfilling ones? *Should* we develop nonhuman minds that might eventually outnumber, outsmart, obsolete and replace us? *Should* we risk loss of control of our civilization? Such decisions must not be delegated to unelected tech leaders. **Powerful AI systems should be developed only once we are confident that their effects will be positive and their risks will be manageable.** This confidence must be well justified and increase with the magnitude of a system's potential effects. OpenAI's [recent statement regarding artificial general intelligence](#), states that *"At some point, it may be important to get independent review before starting to train future systems, and for the most advanced efforts to agree to limit the rate of growth of compute used for creating new models."* We agree. That point is now.

Therefore, we call on all AI labs to immediately pause for at least 6 months the training of AI systems more powerful than GPT-4. This pause should be public and verifiable, and include all key actors. If such a pause cannot be enacted quickly, governments should step in and institute a moratorium.

AI labs and independent experts should use this pause to jointly develop and implement a set of shared safety protocols for advanced AI design and development that are rigorously audited and overseen by independent outside experts. These protocols should ensure that systems adhering to them are safe beyond a

reasonable doubt [4] This does not mean a pause on AI development in general.

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AI research and development should be refocused on making today's powerful, state-of-the-art systems more accurate, safe, interpretable, transparent, robust, aligned, trustworthy, and loyal.

In parallel, AI developers must work with policymakers to dramatically accelerate development of robust AI governance systems. These should at a minimum include: new and capable regulatory authorities dedicated to AI; oversight and tracking of highly capable AI systems and large pools of computational capability; provenance and watermarking systems to help distinguish real from synthetic and to track model leaks; a robust auditing and certification ecosystem; liability for AI-caused harm; robust public funding for technical AI safety research; and well-resourced institutions for coping with the dramatic economic and political disruptions (especially to democracy) that AI will cause.

Humanity can enjoy a flourishing future with AI. Having succeeded in creating powerful AI systems, we can now enjoy an "AI summer" in which we reap the rewards, engineer these systems for the clear benefit of all, and give society a chance to adapt. Society has hit pause on other technologies with potentially catastrophic effects on society.^[5] We can do so here. Let's enjoy a long AI summer, not rush unprepared into a fall.

We have prepared some FAQs in response to questions and discussion in the media and elsewhere. You can find them [here](#).

In addition to this open letter, we have published a set of policy recommendations which can be found [here](#):

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April 5, 2023



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March 29, 2023



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March 29, 2023



Elon Musk and Others Call for Pause



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Signatories

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Yoshua Bengio, Founder and Scientific Director at Mila, Turing Prize winner and professor at University of Montreal

Stuart Russell, Berkeley, Professor of Computer Science, director of the Center for Intelligent Systems, and co-author of the standard textbook "Artificial Intelligence: a Modern Approach"

Elon Musk, CEO of SpaceX, Tesla & Twitter

Steve Wozniak, Co-founder, Apple

Yuval Noah Harari, Author and Professor, Hebrew University of Jerusalem.

Emad Mostaque, CEO, Stability AI

Andrew Yang, Forward Party, Co-Chair, Presidential Candidate 2020, NYT Bestselling Author, Presidential Ambassador of Global Entrepreneurship

John J Hopfield, Princeton University, Professor Emeritus, inventor of associative neural networks

Valerie Pisano, President & CEO, MILA

Connor Leahy, CEO, Conjecture

Jaan Tallinn, Co-Founder of Skype, Centre for the Study of Existential Risk, Future of Life Institute

Evan Sharp, Co-Founder, Pinterest

Chris Larsen, Co-Founder, Ripple

Craig Peters, CEO, Getty Images

Max Tegmark, MIT Center for Artificial Intelligence & Fundamental Interactions,

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Tristan Harris, Executive Director, Center for Humane Technology

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Emilia Javorsky, Physician-Scientist & Director, Future of Life Institute

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Gary Marcus, New York University, AI researcher, Professor Emeritus

Vincent Conitzer, Carnegie Mellon University and University of Oxford, Professor of Computer Science, Director of Foundations of Cooperative AI Lab, Head of Technical AI Engagement at the Institute for Ethics in AI, Presidential Early Career Award in Science and Engineering, Computers and Thought Award, Social Choice and Welfare Prize, Guggenheim Fellow, Sloan Fellow, ACM Fellow, AAAI Fellow, ACM/SIGAI Autonomous Agents Research Award

Huw Price, University of Cambridge, Emeritus Bertrand Russell Professor of Philosophy, FBA, FAHA, co-founder of the Cambridge Centre for Existential Risk

Zachary Kenton, DeepMind, Senior Research Scientist

Ramana Kumar, DeepMind, Research Scientist

Jeff Orlowski-Yang, The Social Dilemma, Director, Three-time Emmy Award Winning Filmmaker

Olle Häggström, Chalmers University of Technology, Professor of mathematical statistics, Member, Royal Swedish Academy of Science

Michael Osborne, University of Oxford, Professor of Machine Learning

Raja Chatila, Sorbonne University, Paris, Professor Emeritus AI, Robotics and Technology Ethics, Fellow, IEEE

Moshe Vardi, Rice University, University Professor, US National Academy of Science, US National Academy of Engineering, American Academy of Arts and Sciences

Adam Smith, Boston University, Professor of Computer Science, Gödel Prize, Kanellakis Prize, Fellow of the ACM

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Academy of Sciences.

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Henry Elkus, Founder & CEO: Helena

Gaétan Marceau Caron, MILA, Quebec AI Institute, Director, Applied Research Team

Peter Asaro, The New School, Associate Professor and Director of Media Studies

Jose H. Orallo, Technical University of Valencia, Leverhulme Centre for the Future of Intelligence, Centre for the Study of Existential Risk, Professor, EurAI Fellow

George Dyson, Unaffiliated, Author of "Darwin Among the Machines" (1997), "Turing's OPEN LETTERS Cathedral" (2012), "Analogia: The Emergence of Technology beyond Programmable Control" (2020).

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Nick Hay, Encultured AI, Co-founder

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Shahar Avin, Centre for the Study of Existential Risk, University of Cambridge, Senior

Research Associate



Signatories

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Solon Angel, AI Entrepreneur, Forbes, World Economic Forum Recognized

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Erik Hoel, Tufts University Professor, author, scientist, Forbes 30 Under 30 in Science, October 25, 2023

Kate Jerome, Children's Book Author/ Cofounder Little Bridges, Award-winning



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